

An Overview of Computing-in-Memory Circuits With DRAM and NVM

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Abstract—Computing-in-memory (CIM) has emerged as an energy-efficient hardware solution for machine learning and AI. While static random access memory (SRAM)-based CIM has been prevalent, growing attention is directed towards leveraging dynamic random access memory (DRAM) and non-volatile memory (NVM) with its unique characteristics such as high-density and non-volatility. This brief reviews the evolving trends in DRAM and NVM-based CIM, which have faced unique challenges that arise from SRAM despite their advantages. For instance, the DRAM cell's density comes with leakage and refresh issues, impacting efficiency and computing accuracy. NVM-CIM faces computing accuracy challenges of resistance-based computation with low signal margins and non-linear characteristics. This tutorial discusses the current status and future directions in DRAM-CIM and NVM-CIM research, which address the abovementioned challenge.

Index Terms—Computing-in-memory, dynamic random access memory, non-volatile memory, magnetic random access memory, resistive random access memory, phase change memory, hardware for artificial intelligence.

I. INTRODUCTION

NOWADAYS, there is a rising surge of interest in Computing-in-memory (CIM) as a potent energy-efficient hardware solution for machine learning (ML) and artificial intelligence (AI) acceleration [1], [2]. CIM involves the integration of high-efficiency computational logic within the memory array, thereby leading to a substantial reduction in the memory and computation energy and a marked enhancement in the energy efficiency of AI/ML applications. Early CIM architecture emerged using NVM with its straightforward mapping [3] and DRAM with its high density [4]. However, in recent years, attention to static random access memory (SRAM)-based implementation has increased rapidly [5], [6], [7], [8], [9], [10], [11], [12], [13], [14]. The rationale behind this trend is rooted in SRAM's characteristic of stably retaining data through static storage

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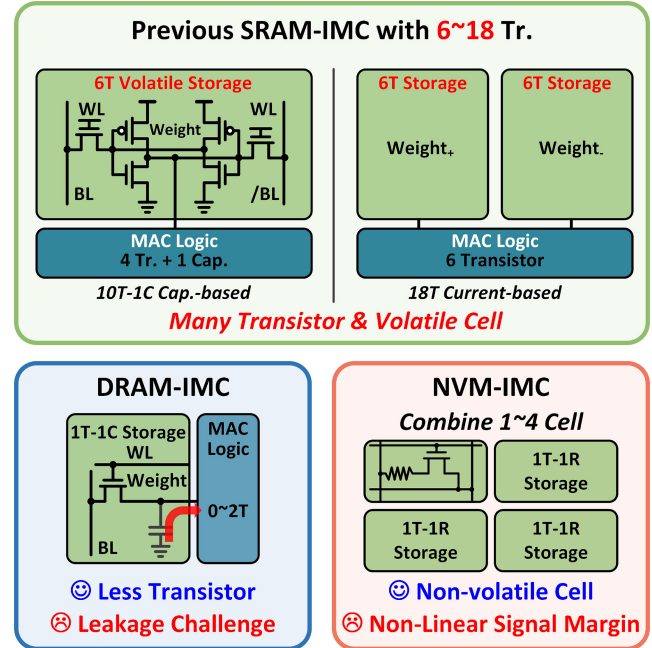


Fig. 1. The Advantage and the Challenge of the DRAM-CIM and NVM-CIM.

utilizing cross-coupled inverter logic. Thanks to stable storage, various computational methods can be used, and their efficiency and accuracy are steadily improved. Moreover, SRAM boasts the advantages of low power consumption for read and write operations, alongside the availability of high-density cells with pushed-rule across various manufacturing processes.

Despite its advantages, SRAM-CIM has inherent drawbacks compared to DRAM-CIM and NVM-CIM that deserve consideration. First, in addition to 6 transistors for the SRAM cell, SRAM-CIM requires additional transistors for computing logic. Therefore, a relatively high transistor count (6 to 18) [5], [6], [7], [8], [9], [10], [11], [12], [13], [14] is required for a unit MAC, and $\times 1.4$ to $\times 2$ [12], [13] more area consumed. Second, SRAM-CIM is confined to storing only 1 bit in a single cell and limits the weight density because of the cross-coupled storage node. Finally, SRAM is unsuitable for applications requiring long-term data storage as it loses data when the power is turned off. This volatility makes it less practical for event-driven applications.

As the field of CIM progresses, these limitations underscore the need to explore alternative memory technologies, and DRAM and NVM-based CIM are being reexamined. DRAM and NVM offer unique advantages and might address some

challenges of the SRAM-CIM. For example, a DRAM cell can be implemented even with a single transistor and capacitor, achieving high memory density. Also, DRAM can be implemented on both DRAM-dedicated processes for higher density and logic processes as embedded DRAM (eDRAM) for logic compatibility and high availability. In addition, NVM can achieve low standby power, high density with multi-level cells, and low system power by eliminating the initial data writing. Therefore, NVM-CIM can provide a highly energy-efficient event-driven system by using power-off mode while maintaining the neural network (NN) model on-chip. In the latest CIM research, implementation using DRAM [15], [16], [17], [18], [19], [20], [21], [22] and NVM [23], [24], [25], [26], [27], [28], [29] is proposed for higher performance and functionality.

Although DRAM and NVM offer advantages in terms of memory density and functionality, the integration of CIM gives rise to fresh challenges. In contrast to the stability of weight data in SRAM-based implementations, the stored data within a cell becomes less stable in DRAM due to leakage and less easy to use in NVM where data is stored as resistance. To handle these new characteristics, new circuits and datapaths are developed to manage the leakage and noise of DRAM, non-linearity, and low signal margin of NVM.

This tutorial will cover the challenges and trends of the computation method in DRAM and NVM-based CIM. Also, the future direction and possible challenges of DRAM and NVM-CIM will be discussed.

II. DRAM-CIM

In the case of DRAM cells, the storage node is implemented using only a capacitor instead of the cross-coupled logic of the SRAM. This design enables the cell with a minimal structure of just one transistor and capacitor. However, the charge on the capacitor can be gradually degraded due to leakage or impacted by noise. However, during the reading process, a sense amplifier detects this charge as 0 or 1, ensuring accurate data transmission from the cell to the computing logic without errors. In the case of DRAM-CIM, where computing logic is integrated within the DRAM memory, data in the cell can be directly used by computing logic inside the cell, facing the challenge of leakage and noise impact.

A. 1T-1C DRAM

The 1T-1C DRAM cell is most broadly used because it achieves the highest memory density with a minimal number of transistors for a cell. Because the design cost of changing the internal structure of the cell array is high, digital processing-in-memory (digital-PIM) or processing-near-memory (PNM) has been most widely studied. The digital-PIM [30], [31] integrates the computing logic after the sense amplifier (SA), and PNM integrates the computing logic on the buffer device of Dual In-line Memory Module (DDIM) [32] or logic die of 3D DRAM [4]. However, the energy efficiency and latency improvement of digital-PIM and PNM is limited since computing logic is not actually placed inside the cell array and SA operates for every data read.

In [33], [34], [35], [36], [37], a method of performing a boolean operation on the bit line (BL) or SA by activating

multiple rows at a time during a read operation was proposed. These approaches use the bitwise operation to support more complex digital operations with primitive boolean operations. Additionally, more bandwidth is available since the computation is performed inside the cell array, and computation efficiency can be further improved than digital-PIM.

Recent research [15], [16] applied analog operation in DRAM CIM to achieve higher efficiency and parallelism. In the [15], the capacitor of the 1T1C cell is used to create the logic necessary to multiply-and-accumulate (MAC) the input activation (IA) and weight (W) of the NN model. First, an analog voltage corresponding to the IA value is generated using charge sharing between cell capacitors and written to the cell. Then, charge sharing is performed between the cells according to the weight value to perform MAC with multi-bit IA and W. Also, [16] adopts the spike neural network (SNN), representing IA as a spike instead of the scalar. Therefore, SNN operation consists of simple *integrate-and-firing* operation instead of *multiply-and-accumulation* operation [16]. implement the integration operation using charge sharing of cell columns and the firing operation as SA operation.

B. Gain-Cell DRAM

The 1T-1C cell uses single-bit lines for read and write operations, resulting in destructive read. Therefore, the data in the cell is destroyed by charge sharing during every read or computation, and SA operation is required to recover it. As a result, the 1T-1C DRAM-CIM consumes considerable power by frequent SA operations. The gain-cell structure is devised to separate the read-bit line (RBL) and write-bit line (WBL) by using an additional transistor. Therefore, although gain cells show lower density than the 1T-1C cell, the non-destructive read path can be used as a computing datapath for less frequent SA operation. Also, DRAM-CIM with gain-cell uses 2-4 transistors for a single cell, which is still lower than that of SRAM-CIM with 6-18 transistors.

The 4T-2C cell proposed in [20] is a pair of 2T-1C gain-cells for complementary storage with signed number representation. Each 2T-1C cell pair performs multiplication of 1-b IA and ternary W, as shown in Fig. 2(a). The driving transistor is connected to the storage node and drives the current to RBL only when its cell value (W or $\overline{W}$) is H and IA on word line (WL) is L . Also, by activating 64 rows in parallel, the multiplication results from each cell can be accumulated with current-based operation. However, the process variation may involve the current amount mismatch among the cells.

To address this issue, as shown in Fig. 2(b), the cell array proposed in [17] removes the row parallelism and integrates the self-detect voltage clipper below every row. After 1'b multiplication by the cell, the clipper logic clips the RBL voltage to the predefined reference voltage. After that, the RBL voltage of every column is accumulated by charge sharing. This operation uses only column parallelism, but the impact of leakage and variation can be eliminated by clipping.

To maintain higher row parallelism while reducing the impact of the variation and leakage, [19] proposed the operation with segmented BL. As shown in Fig. 2(c), the RBL is divided into multiple row segments with a transmission gate instead of sharing the charge among the columns. Each segment's RBL voltage is accumulated with charge sharing after

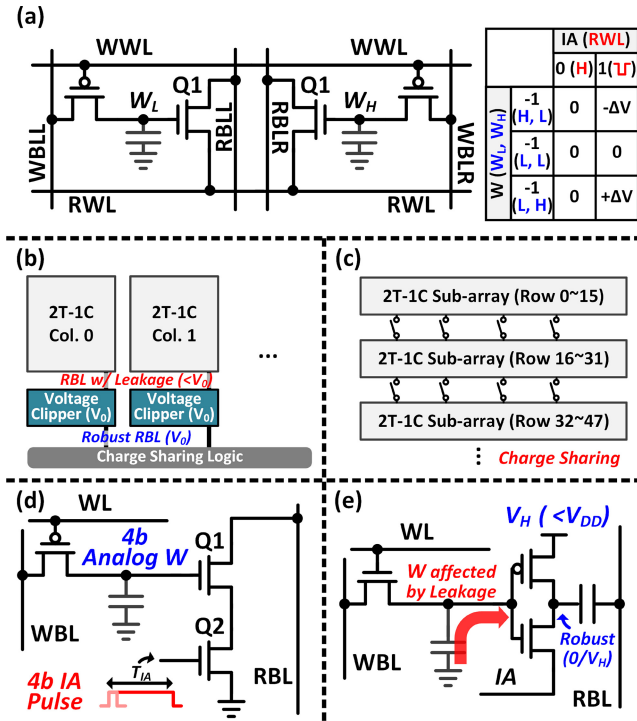


Fig. 2. Operation of the DRAM-CIM with Gain-Cell. (a) 4T-2C Current-based (b) 2T-1C Charge Sharing-based (BL) (c) 2T-1C Charge Sharing-based (sub-BL) (d) 3T-1C Current-based (e) 3T-2C Capacitive Coupling-based.

one row per segment performs 1-b multiplication. Therefore, all segments can be operated in parallel. Also, [19] discovered that most computation errors, such as leakage-induced error and coupling noise from control and word line, are common-mode errors. Therefore, [19] uses one more cell array as a reference cell array to use as the reference voltage of analog-digital converter (ADC) to remove the common-mode error.

3T cell designs [18], [21], [22] have been proposed for more functionality or parallelism than 2T cells. The 3T gain-cell proposed in [18] stores 4-bit weight in a single cell as an analog voltage on a storage capacitor. Also, two transistors are serially connected to the read bit line, as shown in Fig. 2(d). The driving transistor (Q1) connected to the storage node enables multi-bit weight operation by changing the current amount depending on the analog voltage on the storage node. Also, the other transistor (Q2) supports multi-bit IA operation by changing the turn-on time with pulse width. As a result, the single-bit cell stores 4b weight and performs analog 4b-4b multiplication. However, to store 4-b analog weight, a large capacitor is required to reduce the impact of leakage and noise and limit cell density. Also, the current-based operation limits the accuracy due to nonlinearity and variation of the transistor.

The 3T-2C cell in Fig. 2(e) achieves higher cell density using its novel leakage-tolerant computing method [21], [22]. They tailor the capacitive coupling-based operation of SRAM-CIM [12] as operation for DRAM to enable high matching and linearity. Usually, the capacitive coupling-based operation consists of the multiplication in cell and accumulation using a capacitor. The 3T-2C cell in [21], [22] separates the voltage domain of multiplication logic in the cell and multiplication can be done in the digital domain. Therefore, the leakage noise on the storage node can not affect the 1-b multiplication and

TABLE I
COLLECTION OF THE DRAM-PIM

Ref.	Process	Cell Type	Operation	Leakage Handling
[15]	65nm	1T-1C	Charge Sharing (Cell)	No reuse (Destructive Read)
[16]	28nm	1T-1C	Charge Sharing (sub-BL)	No reuse (Destructive Read)
[17]	65nm	2T-1C	Charge Sharing (BL)	Voltage Clipper
[18]	65nm	3T-1C	Current	Cell Capacitor Boosting
[19]	28nm	2T-1C	Current + Charge Sharing (sub-BL)	Reference Cell Array
[20]	65nm	4T-2C	Current	Cell Capacitor Boosting
[21, 22]	28nm	3T-2C	Capacitive Coupling	Leakage Tolerant Computing

accumulation results. Finally, the storage capacitor does not need to be excessively large, resulting in high cell density.

III. NVM-CIM

NVM, such as resistive random access memory (ReRAM), magnetic random access memory (MRAM), and phase-change memory (PCM) represent data as a resistance in the cell. RRAM, MRAM, and PCM differ in mechanisms and performance characteristics. ReRAM uses a metal-insulator-metal structure with metal oxide, MRAM uses manipulation of the magnetization direction of a magnetic tunnel junction (MTJ) and PCM exploits the reversible phase change between amorphous and crystalline states of a chalcogenide material. NVM brings unique advantages to CIM, non-volatility, which maintains data when power is off, and higher density compared to SRAM [38]. Due to these advantages, several NVM-CIM architectures [39], [40] have been studied since early times. Additionally, the latest NVM-CIMs [23], [24], [25], [26], [27], [28], [29] support matrix-vector-multiplication (MVM) operations to demonstrate NN inference. However, regarding NVM-CIM for NN inference, the current-based operation is inevitable since NVM represents data as resistance. Therefore, the static current during the computation, non-linear voltage characteristics, and limited sensing margin become the challenge of the NVM-CIM. This section summarizes recent progress in circuits and datapaths of NVM-CIM that address these challenges.

A. Current-Based Operation With Complementary Cell

The operation with multiple cells has been proposed to overcome the effect of non-linearity and low signal margin on computation accuracy.

The XNOR-RRAM proposed in [26] uses a pair of 1T-1R ReRAM cells for binarized (+1 and -1) IA and W. Each 1T-1C cell pair performs XNOR operation with 1-b IA and 1-b W, as shown in Fig. 3(a). Depending on the weight value, one cell is set to LRS and the other to HRS. Also, only one row of the two rows in the cell pair is activated according to the IA value for XNOR operation, and all pairs are activated in parallel for the accumulation.

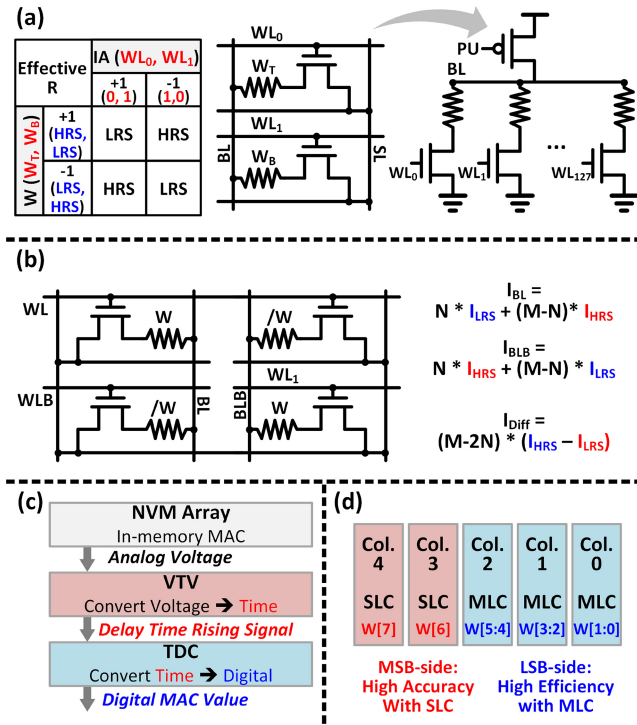


Fig. 3. Operation of NVM (a) 2T-2R XNOR cell (b) 4T-4R Dual Complementary Cell (c) Time-space Readout (d) SLC-MLC hybrid IMC.

Therefore, the voltage of the BL is determined by voltage division with the cell resistance and the pull-up (PU) transistor resistance, as shown in Fig. 3(a). However, because of the non-ideal resistance of BL and pull-up logic, the transfer function of BL voltage becomes non-linear. Therefore, [26] uses flash ADC with non-linear reference voltage for sensing non-linear voltage transfer function with linear quantization. Also, the [27], [28] also adopts the XNOR-based operation for MRAM-CIM. However, [27] extend the 1-b XNOR operation as the multi-bit operation with 1-b input-vector and 4-b matrix. Therefore, for the multi-bit operation with high linearity, the current-based operation with the current combiner feeds the 4-column output to a single ADC simultaneously. Also, [28] uses refined bit-cell to support XNOR operation with enhanced readout margin.

As shown in Fig. 3(b), [29] proposed the 4T-4R cell with dual complementary coding to overcome the non-linearity of the voltage transfer function. In addition to the previous 2T-2R cell that uses complementary WL, the complementary BL was also used and four cells operate as a pair. Therefore, the number of activated LRS and HRS in BL and BLB was always the same and the differential voltage of the BL pair became linear. Therefore, unlike [26] with flash ADC, linear output sensing is possible even with the successive approximation (SAR)-ADC. In addition, the 4T-4R cell in [29] can be reconfigured into the two 2T-2R cells with higher efficiency and lower accuracy. Therefore, it can adaptively reconfigure the mode depending on the accuracy requirements of the NN model for higher efficiency.

B. Time-Based Operation

To overcome the large DC-current flow during voltage-based operation of NVM-CIM, [23] proposed

time-space-based in-memory computing. As shown in Fig. 3(c), [23] uses the series of voltage-to-time converter (VTC) and time-to-digital converter (TDC) after the in-memory operation. After precharging the parasitic capacitor of BL, the current flowed through the cell with the computation results, and discharge latency was sensed. Therefore, no DC-current occurs during the operation, enabling higher efficiency.

C. Hybrid Method

In addition, the hybrid method uses several computational methods together, using a trade-off between computational accuracy and efficiency.

As shown in Fig. 3(d), [24], [25] utilizes both single-level cell (SLC) and multi-level cell (MLC) of the ReRAM and PCM. The SLC and MLC have a trade-off between memory density, efficiency, and signal margin. The MLC achieves higher weight density and efficiency with the multi-bit data in a single cell but shows a lower signal margin. Therefore, [24] implements hybrid operation with SLC and MLC of PCM by storing the upper 2-bit of 8-bit weight in the SLC cell for better signal margin and storing the remaining lower 6-bit in the MLC cells for higher efficiency. Also, [25] uses ReRAM-CIM and supports reconfigurable SLC-MLC-hybrid modes. For example, the number of bits stored in the MLC cell can be configured among 0, 2, 4, 6, and 8 bits. Therefore, depending on the requirement of the NN model, the weight density and computing efficiency can be balanced with computing accuracy.

Reference [25] utilizes both CIM and near-memory-computing (NMC) in a single macro. The NVM-NMC eliminates the multiple-row activation and places the computing unit below the cell array to ensure accurate computation while sacrificing throughput and efficiency. Therefore, with the trade-off between computing accuracy and efficiency, CIM and NMC can be adaptively used depending on the accuracy requirements of each layer of NN.

IV. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

Numerous studies have proposed methods to address the unique challenges arising from computation within DRAM and NVM, such as the impact of leakage and non-linear signal margin. Despite the recent advancements, the field still faces certain challenges. These challenges highlight the nature of DRAM and NVM-CIM and pave the research direction for future research.

1) Process limitations: NVM has limited processes available depending on the memory type, and the circuit for eDRAM or DRAM-CIM may not apply to each other's process. Also, the density of the custom CIM cell is quite lower than the in-fab cell array.

2) Refreshing DRAM causes energy and throughput overhead in addition to the accuracy and density challenges. Also, when expanding the design to a large scale, refresh control can become challenging to handle computation and memory access together.

3) The write operation of NVM requires high voltage and high latency. Also in some cases applying to CIM, the interactive write-read-verify and multiple voltage sources or pulse width modulation are required

TABLE II
COLLECTION OF THE NVM-PIM

Ref.	Memory	Process	Cell Type	Leakage Handling
[23]	ReRAM	22nm	1T-1R	Time-space Readout (Current-Time-Digital)
[24]	PCM	40nm	1T-1R	SLC-MLC Hybrid
[25]	ReRAM	22nm	1T-1R	SLC-MLC Hybrid CIM-NMC Hybrid
[26]	ReRAM	90nm	2T-2R	Nonlinear Voltage Sensing with Flash ADC
[27]	MRAM	22nm	2T-2R	Current Sensing with IDC
[28]	MRAM	28nm	2T-2R	Refined Bit-cell
[29]	MRAM	28nm	4T-4R	Dual Complementary Code for Linear Voltage

To address these challenges, following future research direction is possible:

1) The CIM with standard high-density bit-cells can maintain high density. The computing logic implemented with peripheral circuits or combining peripheral circuits with cells can be a feasible solution.

2) To compensate for the shortcomings of DRAM-CIM or NVM-CIM, SRAM or digital architecture can be used together.

3) The lightweight algorithms to reduce write and refresh or compiler for efficient model mapping can be co-designed.

V. CONCLUSION

This tutorial has covered computing circuits for CIM, specifically focusing on utilizing dynamic random access memory (DRAM) and non-volatile memory (NVM). SRAM-based CIM has dominated previously, benefiting from stable data storage and low power usage, but limitations in transistor count and volatility have driven exploration into DRAM and NVM as promising alternatives for CIM. The continual advancement of DRAM-CIM and NVM-CIM are overcoming challenges and enhancing energy efficiency and functionality. As the field progresses, it is expected that DRAM and NVM-CIM will play key roles in energy-efficient CIM hardware for AI/ML applications.

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